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Some other types: **a) Haplostemonous**: stamens are uniseriate and equal in number to the petals and opposite the sepals (antisepalous)

b) Obhaplostemonous: Stamens are uniseriate, number equal to petals and opposite the petals (antipetalous)

c) Diplostemonous: Stamens are biseriate, outer antisepalous, inner antipetalous. Example: *Murraya*.

d) Obdiplostemonous: Stamens are biseriate, outer antipetalous, inner antisepalous. Example: Caryophyllaceae.

e) Polystemonous: Numerous stamens are normally many more than the number of petals.



c d e

2. Dithecal: It is a typical type, having two lobes with four microsporangia. They are butterfly shaped in cross section. Example: solanaceae.

4.4.5 Anther attachment

1. Basifixed:(Innate) Base of anther is attached to the tip of filament. Example: *Brassica, Datura*.

2. Dorsifixed: Apex of filament is attached to the dorsal side of the anther. Example: *Citrus, Hibiscus*.

3. Versatile: Filament is attached to the anther at midpoint. Example: Grasses.

4. Adnate: Filament is continued from the base to the apex of anther. Example: *Verbena, Ranunculus, Nelumbo*

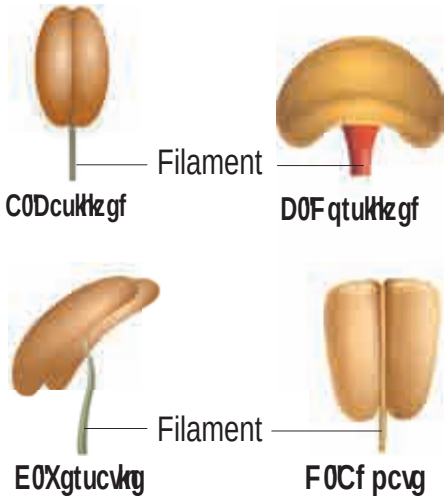


Figure 4.29: Anther attachment

4.4.6 Anther dehiscence

It refers to opening of anther to disperse pollen grains.

1. Longitudinal: Anther dehisces along a suture parallel to long axis of each anther lobe. Example: *Datura*, chinarose, cotton.

2. Transverse: Anther dehisces at right angles to the long axis of anther lobe. Example: Malvaceae.

3. Poricidal: Anther dehisces through pores at one end of the thecae. Example: Ericaceae, *Solanum*, potato, brinjal, *Cassia*.

4. Valvular: Anther dehisces through a pore covered by a flap of tissue. Example: Lauraceae, *Cinnamomum*.

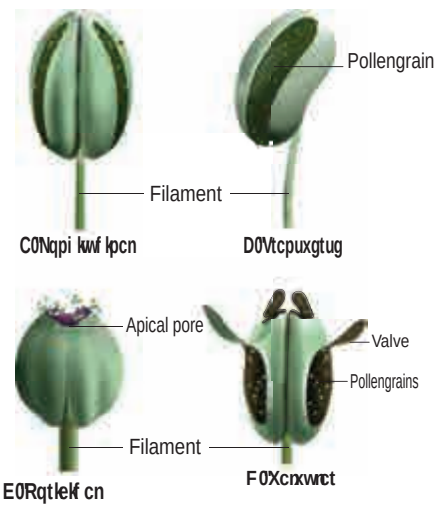


Figure 4.30: Anther dehiscence